MIL-W-19882A(SHIPS)
28 October 1960
SUPERSEDING
MIL-W-19882(SHIPS)
1 May 1957

MILITARY SPECIFICATION

WINCHES, DRUM, POWER OPERATED, BATHYTHERMOGRAPH

1. SCOPE

1.1 This specification covers drum type, power operated, deck mounted bathythermograph winches suitable for controlling the lowering and retrieving of bathythermograph instruments from a surface vessel underway at speeds up to 18 knots.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
TT-P-664 - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-D-963 - Drawings, Production, Procedure for Procurement of.
MIL-C-2174 - Controllers, Motor Starters, and Master Switches Direct-Current Naval Shipboard Use.
MIL-C-2212 - Controllers, Motor Starters, and Master Switches Alternating-Current, Naval Shipboard Use.
MIL-P-3184 - Preparation for Delivery of Capstans, Winches, Windlasses and Associated Repair Parts.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-11796 - Corrosion Preventive, Petrolatum, Hot Application.
MIL-P-15024 - Plates, Identification Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-P-15130 - Paint, Outside, Haze-Gray, No. 27 (Formula No. 5H) (for Naval Shipboard Use).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-B-16392 - Brakes, Electric, Naval Shipboard.
MIL-M-17060 - Motors; Alternating-Current, Integral HP (Shipboard Use).
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard.

STANDARDS

MILITARY

JAN-STD-19 - Welding Symbols.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION

(Application for copies should be addressed to the American Gear Manufacturer's Association, Empire Building, Pittsburgh 22, Pa.)

3. REQUIREMENTS

3.1 Preproduction unit. - One unit of each design shall be subjected to the test of 4.4 and 4.5 prior to production.

3.2 Material. - Material shall be suitable for the purpose intended. The use of material, for mechanical

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items, may be in accordance with either commercial or Government specifications.

3.2.1 If commercial specifications are used, the bill of material shall list both the commercial specification and the Government specification of an equivalent material to be used for replacement purposes, and the nominal chemical and physical characteristics shall be listed on drawings for materials obtained in accordance with commercial specifications.

3.2.2 Gray cast iron shall not be used for any part. Nodular iron may be used for unstressed parts after specific approval by the procuring activity.

3.2.3 Radiographing of steel castings will not be required.

3.2.4 The material shall be magnetic or nonmagnetic as specified (see 6.1). Material with a permeability of less than 1.08 before fabrication and no greater than 2.0 on the finished item shall be considered nonmagnetic. Whenever possible, preference shall be given to nonmagnetic material having the lowest conductivity (electrical). Parts which will have a small total volume, such as rolling contact bearings, may be supplied of ferro magnetic material when the substitution of nonmagnetic material would require extensive development work, excessive item costs, or undue change in manufacturing processes.

3.3 Winch. -

3.3.1 General requirements. - The winch shall consist of a single drum with necessary reduction gearing, clutch, spooling device, revolution counter, electric motor, brake and control equipment, hand crank and all operating accessories. A.c. motor driven units shall be self contained, including control equipment, on a bedplate suitable for securing the equipment permanently to the weather deck. D.c. motor driven units shall have separately mounted controller and resistors. D.c. controller and resistors shall be designed for installation below deck. The winch bedplate shall be of welded construction. The motor shall be located directly underneath the drum. All switches and control handles shall be covered or fabricated with a durable material that will not permit the operator's hand to freeze to it during extreme cold weather operation. Assembly and disassembly of the unit shall not require any special tools. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.) The winches shall be constructed so that all electrical components are interchangeable. (The wire rope will not be supplied by the winch manufacturer, however, for information purposes the wire rope to be used with this winch will be 3/32 inch diameter 7 by 7 corrosion resisting steel wire rope with a breaking strength of 920 pounds.)

3.3.1.1 Welding. - Full use shall be made of welded construction to obtain lightest weight. Welded fabrications shall be stress relieved. Full description of welding procedure shall be given using welding symbols in conformance with Standard JAN-STD-19.

3.3.1.2 Enclosure. - Design of the winch shall be such as to eliminate hazards of parts becoming inoperative due to frequent soaking by salt spray, green seas and extreme atmospheric conditions occuring with the winch in an exposed position on the weather deck of the ship. The winch shall be suitable for operation between ambient temperatures of minus 10° Fahrenheit (F.) and 120°F. and relative humidities of 0 to 100 percent.

3.3.1.3 Weight. - The unit shall be as compact and light as is consistent with adequate strength and rigidity. Overall dimensions in each direction, shall not exceed 24 inches by 31 inches by 36 inches in height and the weight shall not exceed 700 pounds exclusive of the d.c. controller and resistors.

3.3.1.4 Fasteners. - Protective means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. The use of set screws shall be avoided. When set screws are used, means shall be provided to prevent their working out.

3.3.2 Drum. - The drum shall have a minimum barrel diameter of 8 inches; maximum width of 6 inches inside of flanges and a flange diameter of 13-3/4 inches. The drum shall be balanced to 1000 r.p.m. Commercial standards shall be used. The drum shall be guarded as necessary to prevent the wire rope from jumping the drum. The drum flanges shall be of adequate thickness to withstand spreading forces (no permanent set) when the full length of wire rope at 26°F., is spooled at the rated line pull and then allowed to increase in temperature to an ambient temperature of 100°F. The wire rope shall lead from the top of the drum. The design for the drum shall be such that the moment of inertia will be a minimum value. Positive means of attaching the wire rope to the drum barrel shall be provided.

3.3.2.1 Clutch. - The winch drum shall be equipped with a friction type clutch having positive disengagement. The clutch shall drive the rated load at rated speed. When the drum is running free and the electric motor brake is set, the clutch shall be able to smoothly stop the winch rated load (see 3.4) at speeds up to 2000 feet per minute (f.p.m.). The clutch shall be able to hold without slip, a pull equal to 300 percent winch rated line pull, when the wire

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rope is on the bare drum. The clutch shall be engaged and disengaged by means of a handwheel having a handle at the rim. The life of the clutch should not be adversely affected when operating as a brake.

3.3.3 Gearing. - Gear drives shall be used throughout. The gears shall be designed in accordance with Publication American Gear Manufacturers Association, Class 1, for 10,000 hours life. All gearing shall be of the spur, helical or herringbone type, shall have machine cut teeth, shall be totally enclosed in an oiltight case and shall be lubricated by an oil bath. Worm gears and bevel gears are not acceptable.

3.3.4 Bearings and bushings. - Antifriction bearings shall be in accordance with Specifications FF-B-171 and FF-B-185, as applicable, and shall be used throughout for all torque transmitting shafts and for the drum shaft. Bronze bushings shall be used on all accessories and accessory mechanisms. Positive means shall be provided to prevent the bushings from turning and cutting off the supply of lubricant.

3.3.5 Lubrication. - Pressure grease fittings in accordance with type I or III of Specification MIL-F-3541 shall be provided for all bearings and bushings not operating in an oil bath. Sliding clutches, unless operating in an oil bath, shall be provided with at least two grease fittings, 180 degrees apart, or other means of assuring that lubricant will be evenly distributed.

3.3.6 Bedplate. - The winch bedplate shall be of adequate strength to maintain alignment without aid from the deck foundation. The foundation bolt holes shall be drilled 1/16-inch undersize for reaming at installation to fit body bound bolts. The design shall eliminate pockets in which water can stand or collect.

3.3.7 Motors. - Unless otherwise specified in the contract or order, the electric motor may be loaded to 125 percent to obtain rated winch output (see 3.4).

3.3.7.1 Electric direct current (d.c.) motors. - Electric d.c. motors shall be in accordance with Specification MIL-M-17413 and the following:

Service - Navy "A".
Ambient temperature - 40°C.
Voltage - 115 volts d.c. or 230 volts d.c. as specified (see 6.1).
Duty - Short time 30 minutes.
Horsepower - as required.
Enclosure - Watertight.
Speed - As required.
Bearings - Ball.
Insulation - Class A or B.
Winding - Compound.
Shunt field - Rated for continuous connection to line voltage.

3.3.7.2 Electric alternating current (a.c.) motor. - Electric a.c. motors shall be in accordance with Specification MIL-M-17060 and the following:

Service - Navy "A".
Ambient temperature - 40°C.
Voltage - 220 volts, 3 phase 60 cycle, a.c. or 440 volts, 3 phase, 60 cycle, a.c. as specified (see 6.1).
Phase characteristics - Squirrel cage inductor.
Horsepower - As required.
Design - B.
Enclosure - Watertight.
Speed - 2 speed (full and 1/4 constant torque).
Bearing - Ball.
Insulation - Class A, B or H.
Radio noise suppression - Not required.

3.3.8 Motor controllers. - Electric motor controllers shall be as specified in 3.3.8.1 and 3.3.8.2.

3.3.8.1 Electric d.c. controllers. - Electric d.c. controllers shall be in accordance with Specification MIL-C-2174 and the following:

Ambient temperature - 50°C.
Voltage - 115 volts, d.c. or 230 volts d.c. as specified (see 6.1).
Duty - Continuous.
Operation - Magnetic.
Type - Resistor.
Function - Motor starting, and speed selection (3 speed minimum with armature shunting in the first speed point).
Performance - Semi-automatic.
Insulation - Class B (class A may be used for coils).
Enclosure - Dripproof.
Protection features - (LVR) and thermal overload.
Master switch - Remote drum type, watertight, spring return to "OFF".

3.3.8.2 Electric a.c. controllers. - Electric a.c. controllers shall be in accordance with Specification MIL-C-2212 and the following:

Ambient temperature - 50°C.
Voltage - 220 volt, 3 phase, 60 cycle, a.c. or 440 volt, 3 phase, 60 cycle, a.c. as specified (see 6.1).
Duty - Continuous.
Operation - Magnetic.
Type - Across-the-line.
Function - Motor starting and speed selection.
Performance - Semi-automatic.

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Insulation - Class B (class A may be used for coils).
Enclosure - Watertight.
Protection features - LVR and thermal overload.
Master switch - Remote drum type, watertight, spring return to "OFF".

3.3.9 Electric brake. - The electric brake shall be as specified in 3.2.9.1 and 3.2.9.2 and shall be capable of stopping and holding 200 percent winch rated load (see 3.4) and of holding a static load of 300 percent.

3.3.9.1 Electric d.c. brakes. - Electric d.c. brakes shall be in accordance with Specification MIL-B-16392 and the following:

Type - Shoe or disc.
Operation - Solenoid.
Ambient temperature - 40°C.
Voltage - 115 volt d.c. or 230 volt d.c. as specified (see 6.1).
Insulation - Class A or B.
Enclosure - Watertight.

3.3.9.2 Electric a.c. brakes. - Electric a.c. brakes shall be in accordance with Specification MIL-B-16392 and the following:

Type - Shoe or disc.
Operation - Solenoid.
Ambient temperature - 40°C.
Voltage - 220 volts, single phase, 60 cycles a.c. or 440 volts, single phase, 60 cycles a.c., as specified (see 6.1).
Insulation - Class A or B.
Enclosure - Watertight.

3.3.10 Revolution counter. - A gauge graduated in feet of wire rope on the drum, shall be provided on the winch. The graduations shall be such that the operator can easily read hundreds of feet and estimate the amount of cable to the nearest 10 feet. The indicator shall show the length of wire rope payed out and shall be readily adjustable to zero.

3.3.11 Hand crank. - Provisions shall be made for quick and convenient operation of the winch by hand in emergencies. A safety type hand crank shall be mounted to the side of the winch where it will be readily accessible.

3.3.12 Spooling device. - A hand operated spooling device shall be provided for spooling the wire rope on the drum. It shall be possible to disconnect and retract the spooling device for paying out the wire rope. Rollers, two vertical and one horizontal, below the wire rope, shall be provided on the spooling device.

3.3.13 Sheave streaming. - One towing block shall be provided with each winch. The towing block shall be provided suitable for mounting on the underside of an 8-foot horizontal boom at right angles to the side of the ship and at an elevation of approximately 4 feet above the winch foundation. Attachment to the boom shall permit cable lead alignment during installation from 15 degrees forward to 15 degrees aft of the boom to suit winch location. The block shall be accurately balanced to orient itself under minimum cable tension to any cable lead to the towed object between angles of 90 degrees forward and 90 degrees aft and 45 degrees inboard and 45 degrees outboard of a perpendicular from the center of the sheave.

3.3.13.1 The sheave tread diameter shall be approximately 3 inches and shall be grooved to accommodate 3/32 inch, 7 by 7 corrosion resisting steel wire rope.

3.3.13.2 The sheave shall be designed for operation at line speeds up to 2000 feet per minute (f.p.m.) and shall be mounted on antifriction "sealed-for-life" bearings. Means shall be provided for replacement of lubricant in the bearings at time of overhaul.

3.3.13.3 The towing block with the sheave shall be as light and compact as is consistent with the strength required. Stresses shall not exceed limitations established in paragraph 3.5.1.

3.4 Performance. - The winch shall be rated 400 pounds single line pull at 90 and 360 f.p.m. D.c. powered winches shall also provide a line speed of 60 to 90 f.p.m. at 40 pounds line pull in the first speed point. The rating of the winch is with the rope on the average layer of the drum. Speed selection shall be obtained electrically. Note: The average layer is that which is half way between the barrel diameter and the top layer with 3000 feet of wire rope on the drum.

3.5 Stress calculations. - A complete set of stress calculations of gears, bearings, shafts, keys, holding down bolts, and brakes shall be included with the first submission of drawings.

3.5.1 Under rated conditions, the combined stress in any part shall not exceed 35 percent of the tensile yield point of the ductile material used.

3.5.1.1 The combined stress in any part of the machine shall not exceed 70 percent of the tensile yield point when the drum is subjected to a pull equal to 300 percent of the rated load with the rope on the bare drum.

3.5.2 In stress calculations, the strength relationship shall be as shown in table I.

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Table I - Strength relationships

Design Strength	Percent of tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160

3.5.2.1 Other values may be used if substantiated by tests and approved by the bureau or agency concerned.

3.6 Identification plates. - Identification plates and other designating working plates shall be in accordance with type C of Specification MIL-P-15024. This identification plate shall also contain the name of the item and the stock number with the words "Stock Number _____". (This stock number may require as many as seventeen spaces).

3.7 Finish. - Prior to shipment, the winches shall be thoroughly cleaned and all parts not intended to remain bright shall be given two coats of primer conforming to Specification JAN-P-735 or Specification TT-P-664. A final coat of paint conforming to Specification MIL-P-15130 shall then be applied. Exposed surfaces that will be inaccessible after installation (undersides of winch stand, and similar items) shall be given three coats of primer conforming to Specification JAN-P-735 or Specification TT-P-664 or heavily coated with a corrosion preventive compound conforming to class 3 of Specification MIL-C-11796.

3.8 Drawings. - All drawings shall conform to Specification MIL-D-963.

3.8.1 Preliminary drawings. - A complete set of subassembly, cross-sectional assembly, general arrangement, electric motor and controller drawings shall be submitted via the government inspector for approval by the bureau or agency concerned. In addition, all electrical drawings shall conform to the requirements of the applicable equipment specifications. Any departures from these specifications with the reasons therefor shall be stated in an accompanying letter. Approval indicates general conformance with specification requirements; however, it does not relieve the contractor of responsibility for meeting all design and operational requirements.

3.8.2 Any work performed by the contractor, and any materials acquired in advance of the approval by the Government of the preliminary assembly and detail drawings and stress calculations shall be at the contractor's risk.

3.8.3 For the purpose of scheduling fabrication and deliveries, the contractor may assume that 21 days will be required by the bureau or agency concerned for action on each drawing or group of drawings submitted or resubmitted for approval.

3.8.3.1 A numbering system shall be used for identification of winch components that will prevent the duplication of piece numbers on assembly drawings, sectional assembly drawings, detail drawings, and repair part listings.

3.8.4 The general arrangement drawings to be included in the first submission of drawings shall contain the following in addition to the assembly views of the unit:

(a) A list of the mechanical onboard repair parts.
(b) An index of all related drawings with the contractor's number, title and bureau or agency number.
(c) A notation of the weight of the unit, motor and the repair parts.
(d) A notation of duty ratings.

3.8.5 Final drawings shall consist of a microfilm of all assembly, subassembly, detail and electrical drawings and shall be in accordance with Specification MIL-D-963.

3.9. Technical manuals. - Technical manuals shall conform to type III of Specification MIL-M-15071 and shall contain all variations of the winch required by the contract. Two preliminary copies of the technical manual shall be forwarded for approval and assignment of the bureau or agency concerned identification number. These preliminary technical manuals shall contain blue prints of all drawings listed hereinafter for inclusion in the final manual. Lists of drawings, repair parts, and special tools furnished and repair parts lists shall be provided where not shown on the drawings or sketches included. The following prints of corrected working drawings reduced for folding in the lengthwise direction only shall also be made a part of the technical manual.

(a) General arrangement drawing.
(b) Cross sectional assembly and subassembly drawings in sufficient number to adequately identify parts.
(c) Bedplate or foundation drawing.
(d) Lubrication charts with lubricants listed according to Government specification number or symbol number.
(e) Operating instructions.
(f) Maintenance instructions.
(g) Electric motor, controller drawings and certification data.

3.9.1 Preliminary technical manuals. - Preliminary (approved) technical manuals as may be required for installation purposes of the preproduction equipment shall be provided at the time of testing.

3.9.2 Operating instructions describing how the bathythermograph instrument is lowered and retrieved is to be included in the final technical manual. A copy not exceeding two pages will be furnished by the bureau or agency concerned at the time the preliminary technical manual is submitted for approval.

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3.10 Repair parts. -

3.10.1 Mechanical. - Unless otherwise specified in the contract or order, each set of mechanical repair parts shall consist of the following items, as applicable, in sufficient quantity for one winch:

(a) One set of bushing type bearings.
(b) One set of oil seals and special packing.
(c) One set of antifriction bearings.
(d) One set of clutch lining and rivets.
(e) One complete set of springs.

3.10.2 Electrical. - Electrical repair parts shall be furnished in accordance with Specifications MIL-B-16392, MIL-M-17060, MIL-C-2212, MIL-M-17413 and MIL-C-2174, as applicable to the electrical items.

3.11 The manufacturer shall forward two copies of general arrangement, assembly and subassembly drawings as approved by the bureau or agency concerned to Ships Parts Control Center, Mechanicsburg, Pennsylvania, via the Government inspector, for assignment of Federal stock numbers. The manufacturer's list of recommended stock repair parts shall be furnished in accordance with Specification MIL-P-15137.

3.12 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contractor or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests. -

4.2.1 Place of tests. -

4.2.1.1 Tests at place of manufacturer. - Preproduction tests shall be conducted on one winch of each design by the contractor, at his own expense, at the place of manufacture. These tests may be witnessed by authorized representatives of the bureau or agency concerned.

4.2.1.1.1 The contractor shall notify the Bureau of Ships, Code 632B, via the Inspector of Naval Material of the place and time when such tests will be conducted at least three weeks prior to commencement of tests. A proposed schedule and outline of tests to be conducted shall be forwarded with the notification.

4.2.1.2 Tests onboard ship. - Preproduction tests shall be conducted onboard ship on one winch of each design. Testing of the equipment onboard ship will require 90 days.

4.2.2 Tests. - Preproduction tests shall be as specified in 4.4. and 4.5. (Wire rope for testing the equipment at the place of manufacture shall be furnished by the manufacturer. A 1/8-inch diameter 7 by 19 wire rope having a breaking strength of 2000 pounds is suggested).

4.2.3 Acceptance or rejection. - Within 30 days after completion of the tests specified in 4.4 and 4.5, the contractor will be notified by the Government inspector whether the preproduction unit has complied with the requirements specified herein by means of a written approval, conditional approval, or disapproval. In the event that the contractor does not receive such notification within such period, and promptly so notifies the Contracting Officer in writing, the contract delivery date shall be extended by the Contracting Officer by the number of days which notification is delayed. In the event of disapproval, the Contractor shall correct all deficiencies before retesting the preproduction unit. Any such retesting shall be conducted in accordance with the requirements and procedures specified in 4.2.

4.2.4 The approved preproduction unit shall serve as a manufacturing standard any may be submitted as a production unit provided that all modifications required to meet the requirements of this specification have been incorporated therein.

4.2.5 In addition to any requirements specified herein or in the contract or order with respect to the performance of the tests, the contractor shall make a record of all data obtained during such tests in such form as is consistent with good engineering practice and shall furnish the cognizant inspector of material three copies of all such data.

4.3 Sampling for lot acceptance. -

4.3.1 Lot. - All winches of the same design offered for delivery at one time under one contract or order shall be considered an inspection lot.

4.3.2 Sampling for visual and dimensional examination. - A random sample of winches shall be selected from each lot offered for delivery in accordance with Standard MIL-STD-105 at inspection level III. The acceptable quality level shall be 1.5 percent defective.

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4.3.3 Sampling for tests. - One sample winch shall be selected at random from each lot offered for delivery and subjected to the tests specified in 4.5.2 through 4.5.4. Each winch shall be tested as specified in 4.5.1.

4.4 Visual and dimensional examination. - Each of the sample winches selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification. Any sample winch containing one or more visual or dimensional defects shall be rejected and if the number of defective winches in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Tests. -

4.5.1 Operation without load. - Each winch shall be run under no load for a period of one-half hour in each direction. No abnormal heating of any part shall occur.

4.5.2 Load test. - Each sample winch shall hoist from a suspended position, rated load at rated speed and shall hoist 150 percent rated load at no specified speed. No abnormal heating of any part shall occur.

4.5.3 Electric braking test. - Each sample winch shall be tested to determine the ability of the electric brake to stop and hold 200 percent winch rated load and to withstand a static pull of 300 percent rated load without creepage.

4.5.4 Clutch. - Each sample winch shall be tested to determine the ability of the clutch to hold a pull equal to 300 percent winch rated load, to stop and hold 200 percent winch rated load, and to smoothly stop rated load between 60 and 2000 f.p.m.

4.5.5 Rejection. - If any winch fails any test the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. -

5.1.1 The equipment, accessories, technical publications and repair parts shall be preserved and packaged by Level A or C; packed by Level A, B or C and marked Level A, B or C as specified (see 6.1), in accordance with Specification MIL-P-3184.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether magnetic or nonmagnetic material is required (see 3.2.4).
(c) Voltage required for electric d.c. motors, controllers and brakes (see 3.3.7.1, 3.3.8.1 and 3.3.9.1).
(d) Voltage required for electric a.c. motors, controllers and brakes (see 3.3.7.2, 3.3.8.2 and 3.3.9.2).
(e) Selection of applicable level of packaging, packing and marking required (see 5.1).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project No. 3950-N007Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0378

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